

class09

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Data Import

```
candy <- read.csv("candy-data.txt", row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
dim(candy)
```

```
[1] 85 12
```

there are 85 different candy types

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Milky Way", ]$winpercent
```

```
[1] 73.09956
```

The winpercent value is 73.09956 of Milky Way.

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

The winpercent value is 76.7686

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

The winepercent value is 49.6535 of a tootsie roll.

```
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

```
library("skimr")
skim(candy)
```

Table 3: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency: numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

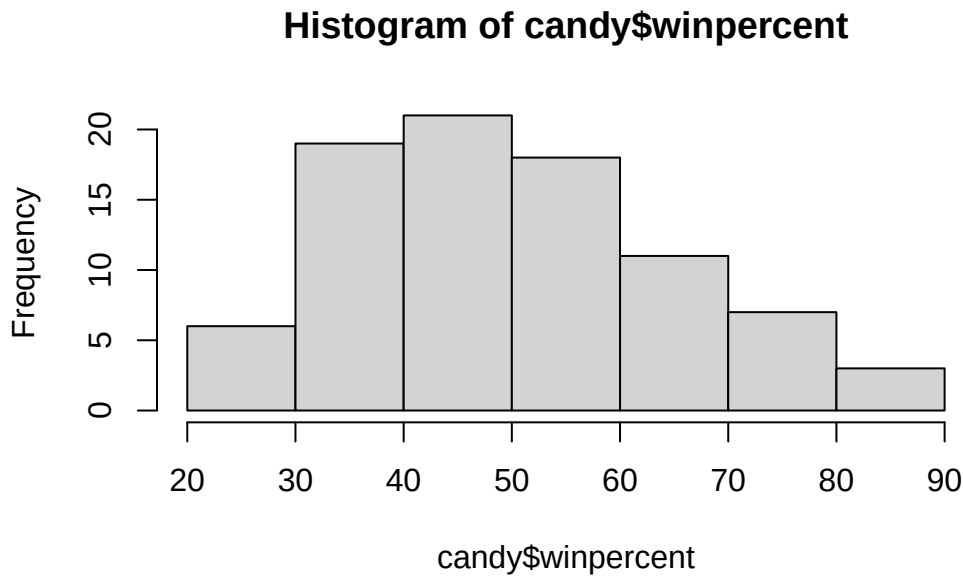
the variables and columns that are on different scales as compared to the majority of the other columns is winpercent this is because most vairables have a value of 0 or 1, and 0 to 1. However, winperecent ranges from values between 1 to 100 which is at a significantly higher scale than the other columns that are limited to values between 0 and 1.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

The zero represents that the candy has no chocolate and the one represents the candy has chocolate

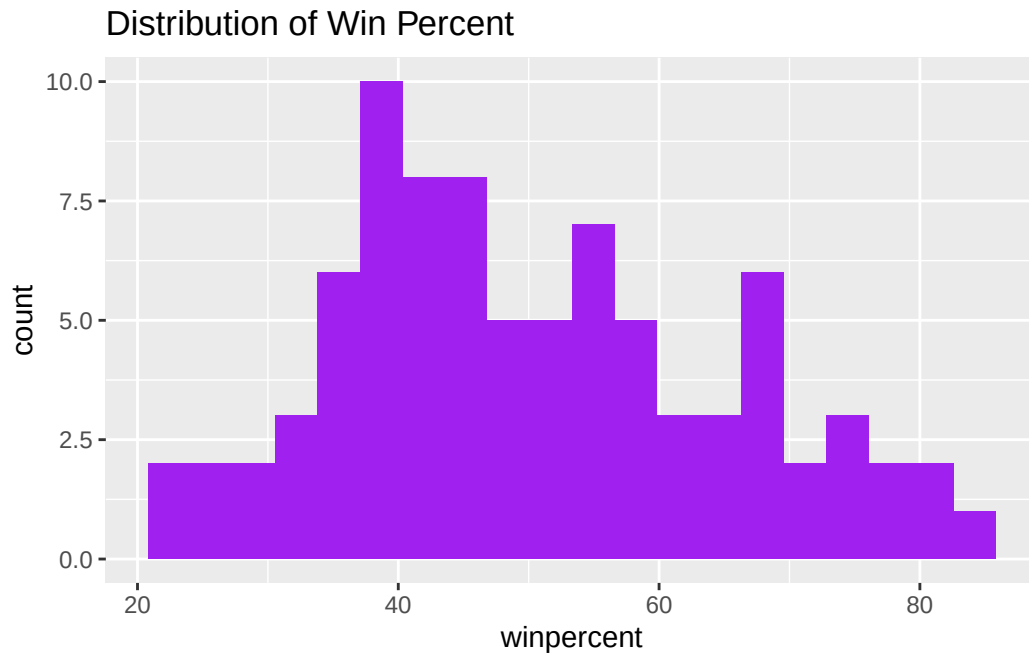
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$winpercent)
```



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins = 20, fill="purple") +
  ggtitle("Distribution of Win Percent")
```



Q9. Is the distribution of winpercent values symmetrical?

No, it seems to be slightly skewed with a slight right skew.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
median(candy$winpercent)
```

```
[1] 47.82975
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The center of the distribution is approximately below 50% as the median is slightly below 50% at 47.83% the mean is at 50.32% which indicates that the center is approximately but more slightly below 50% as the median is below 50% and it represents the center of the data.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
fruit.ind <- as.logical(candy$fruity)
fruit.candy <- candy[fruit.ind,]
fruit.win <- fruit.candy$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

Chocolate candy is ranked higher than fruit candy at 60.92153 and fruity candy is ranked 44.11974.

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

These are significantly significant because the two sample t-test p-values are less than 0.05 at 2.871e-08.

Q13. What are the five least liked candy types in this set?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  arrange(winpercent) |>
  select(winpercent) |>
  head(5)
```

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

```
x <- c(5,10,1)
sort(x)
```

```
[1] 1 5 10
```

```
head(candy[order(candy$winpercent), ])
```

	chocolate	fruity	caramel	peanuty	almondy	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0
Root Beer Barrels	0	0	0		0	0

	crispedrice	wafer	hard	bar	pluribus	sugarpercent	pricepercent
Nik L Nip		0	0	0	1	0.197	0.976
Boston Baked Beans		0	0	0	1	0.313	0.511
Chiclets		0	0	0	1	0.046	0.325
Super Bubble		0	0	0	0	0.162	0.116
Jawbusters		0	1	0	1	0.093	0.511
Root Beer Barrels		0	1	0	1	0.732	0.069
	winpercent						
Nik L Nip	22.44534						
Boston Baked Beans	23.41782						
Chiclets	24.52499						
Super Bubble	27.30386						
Jawbusters	28.12744						
Root Beer Barrels	29.70369						

The five least liked candy types in this set are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters.

Q14. What are the top 5 all time favorite candy types out of this set?

```
library(dplyr)

candy |>
  arrange(desc(winpercent)) |>
  select(winpercent) |>
  head(5)
```

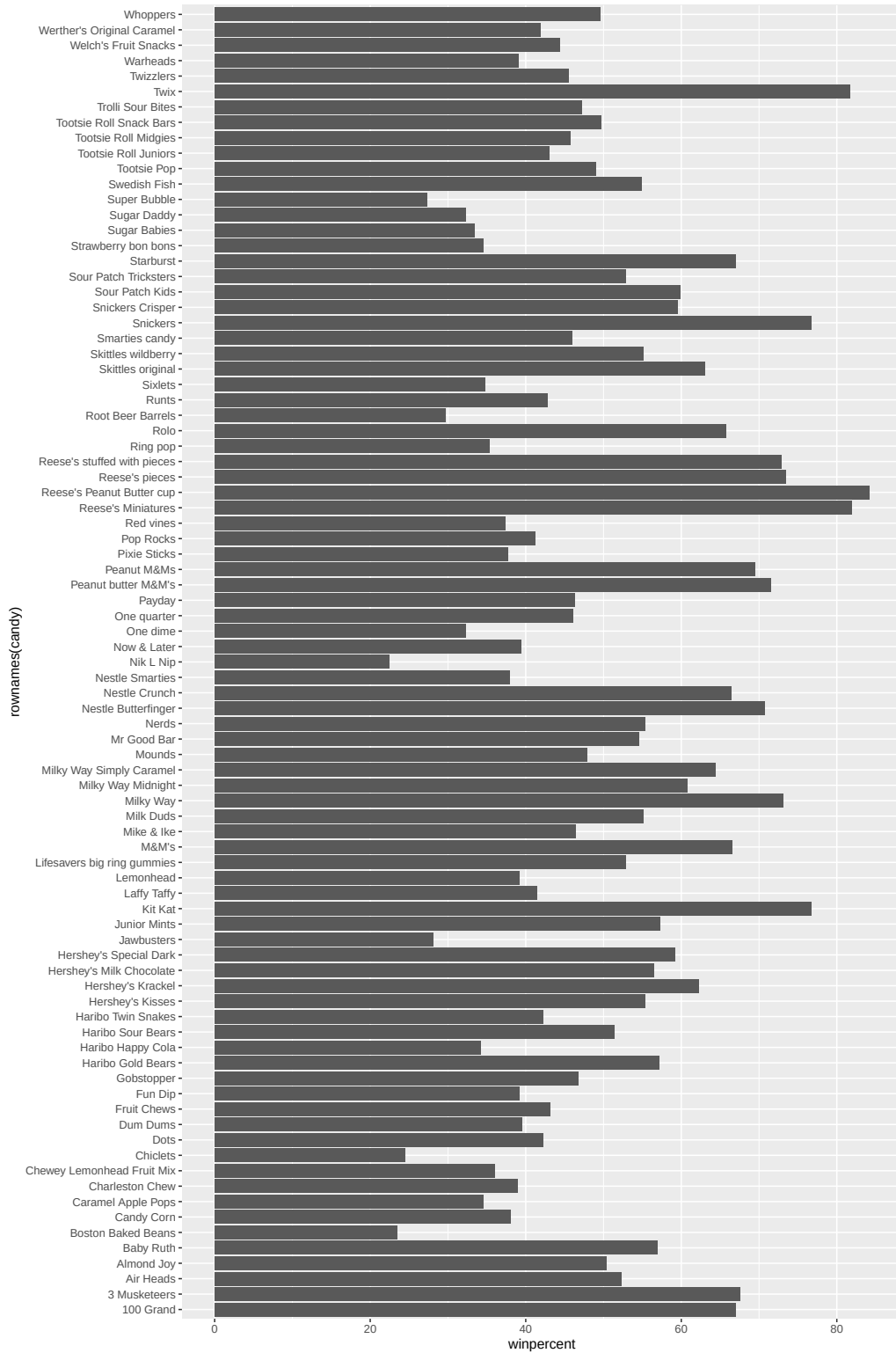
	winpercent
Reese's Peanut Butter cup	84.18029
Reese's Miniatures	81.86626
Twix	81.64291
Kit Kat	76.76860
Snickers	76.67378

The top 5 all time favorite candy are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

Barplot and code of candy rankings are found below:

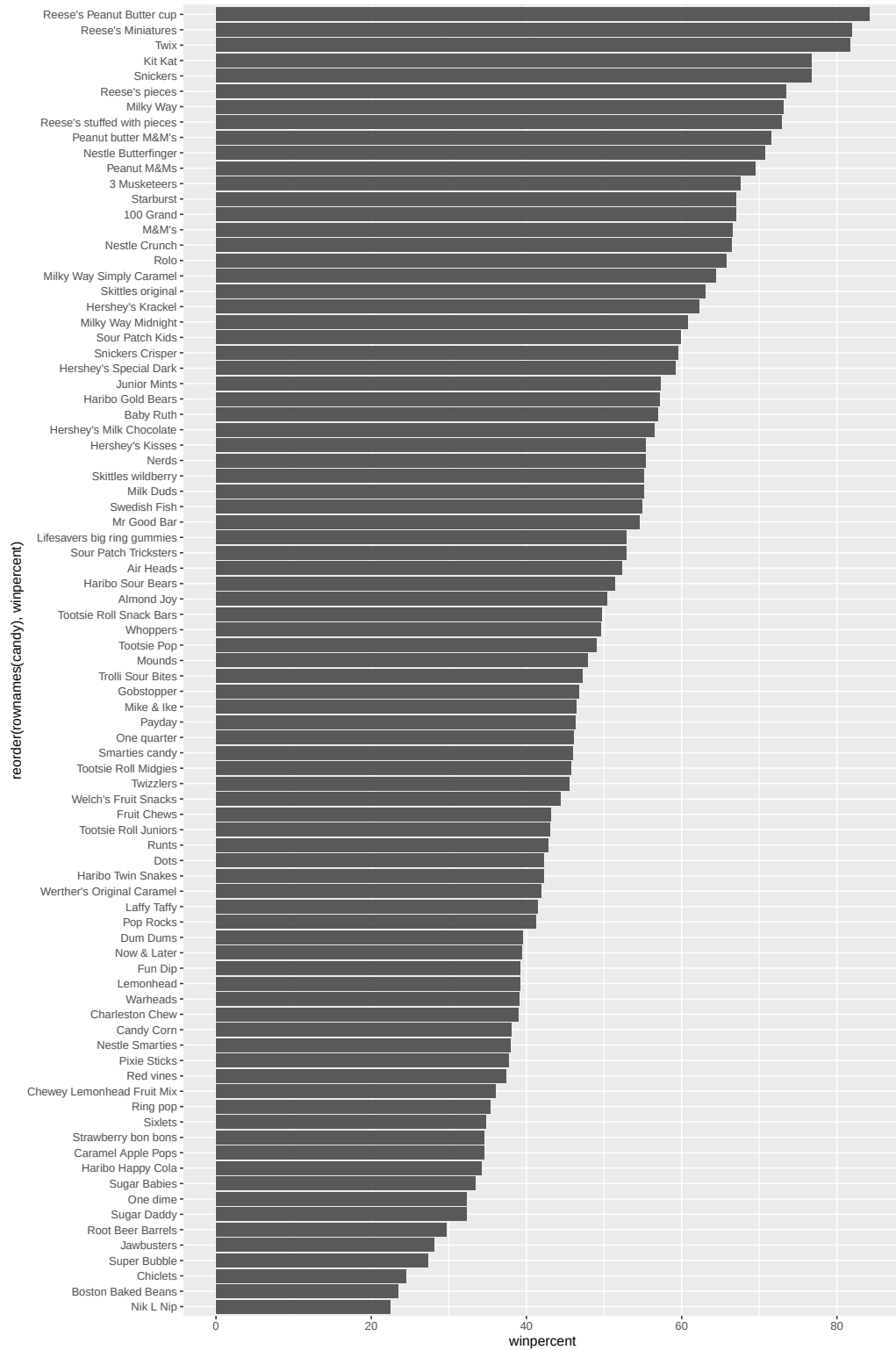
```
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```



16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

With the `reorder()` function:

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



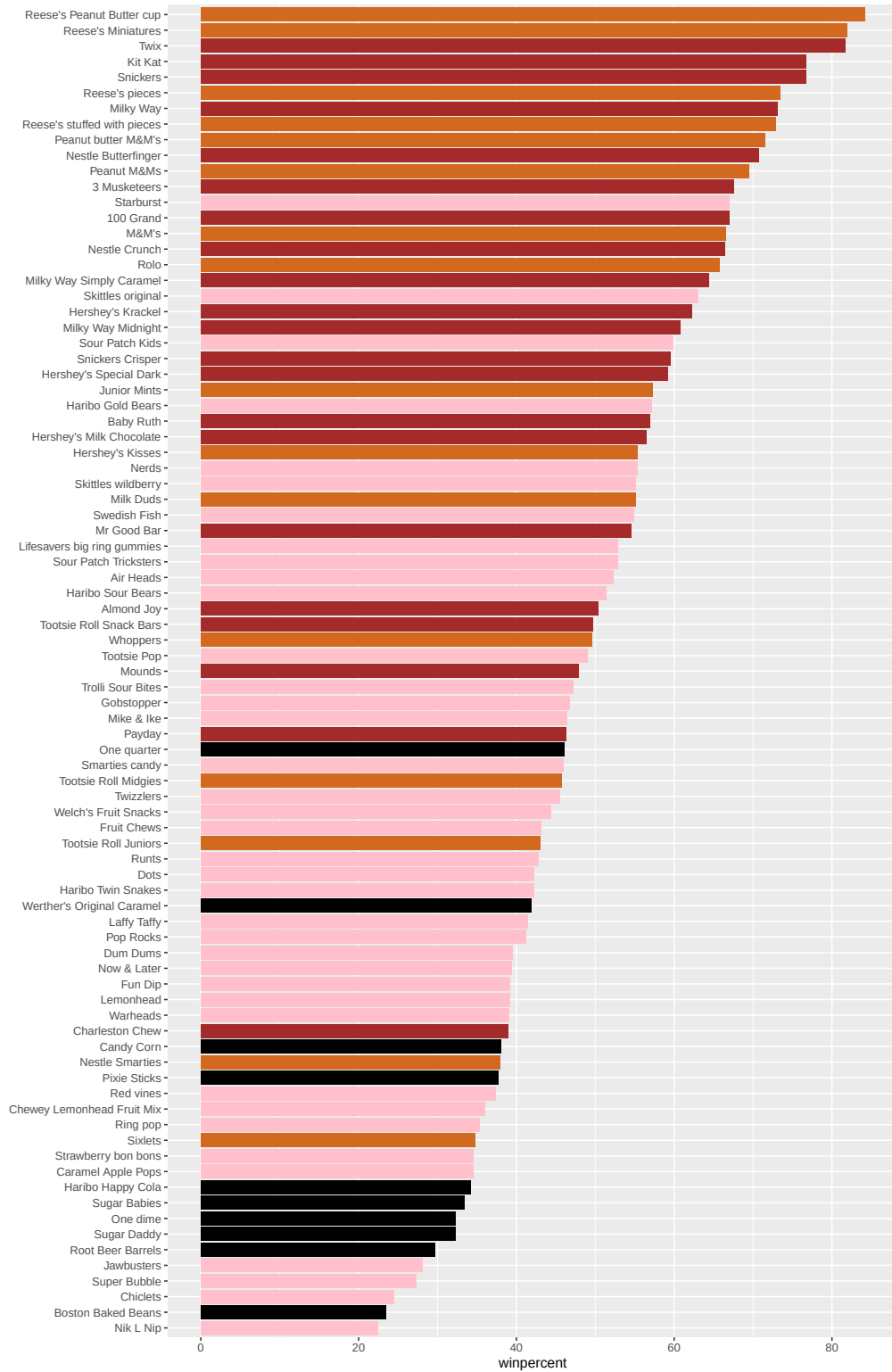
```

mycols <- rep("black", nrow(candy))

#Chololate Candy
mycols[as.logical(candy$chocolate)] <- "chocolate"
#fruity candy in pink
mycols[as.logical(candy$fruity)] <- "pink"
#candy bars un brown
mycols[as.logical(candy$bar)] <- "brown"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=mycols) +
  ylab("")

```



Q17. What is the worst ranked chocolate candy?

sixlets are the worst ranked chocolate candy

Q18. What is the best ranked fruity candy?

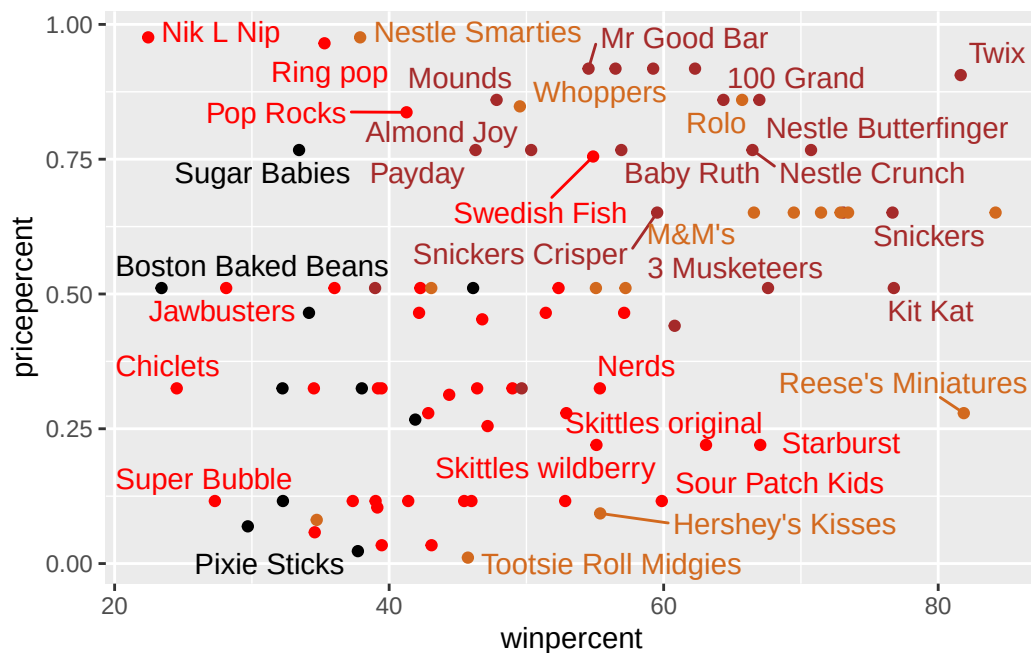
The best ranked fruity candy is starburst

Taking a look at pricepercent

```
# fruity candy in pink
mycols [as.logical(candy$fruity) ] <- "red" #"pink"
```

```
library(ggrepel)

ggplot(candy) +
  aes(winpercent,
      pricepercent,
      label=rownames(candy)) +
  geom_point(col=mycols) +
  geom_text_repel(col=mycols, max.overlaps = )
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's minatures is the highest ranked in terms of winpercent for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The top 5 most expensive candy types are Nik L Nip, Nestle Smarties, Ring Pop, Hersheys Krackel, and Hersheys Milk Chocolate. The least popular of this set is Nik L Nip with a lower winpercent 22.44534.

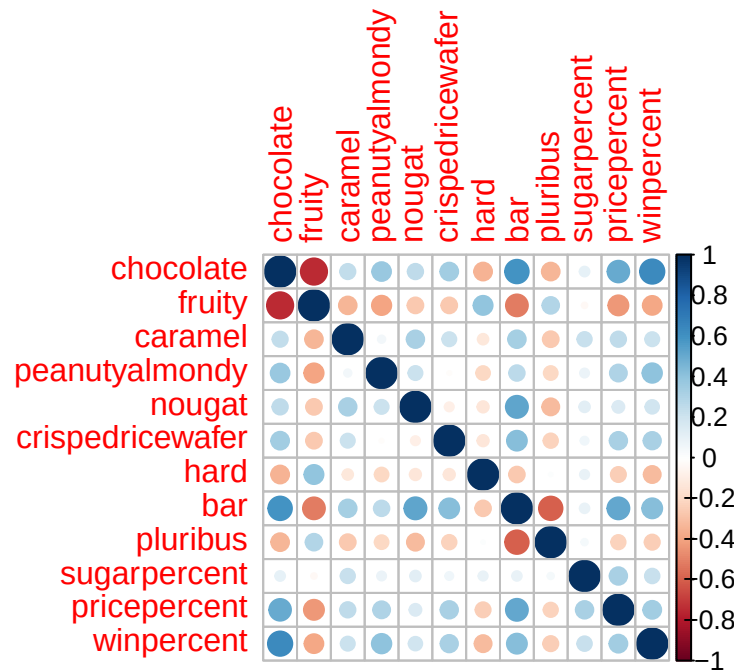
Exploring the correleation structure

We can calculate the pair-wise correlation of all our columns

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruit are the most anti-correlated.

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

I predict that chocolate and fruity candy will feature heavily in the PC1 due to their strong opposing correlation.

PCA

In this case we want to be sure to set `scale=TRUE` because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale =TRUE)
summary (pca)
```

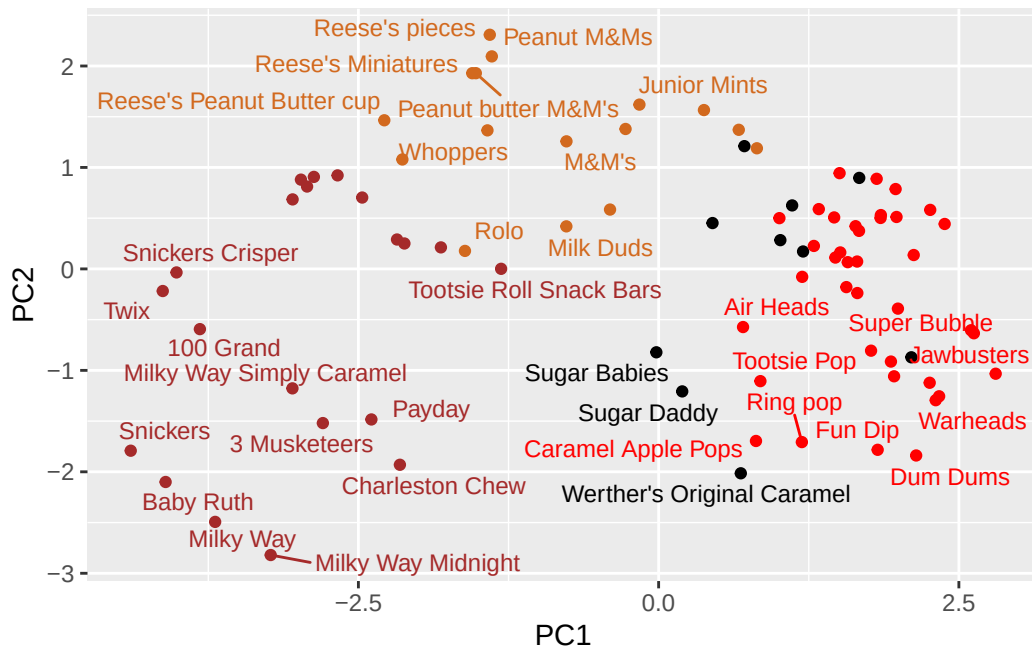
Importance of components:

PC1 PC2 PC3 PC4 PC5 PC6 PC7

Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369
	PC8	PC9	PC10	PC11	PC12		
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760		
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317		
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000		

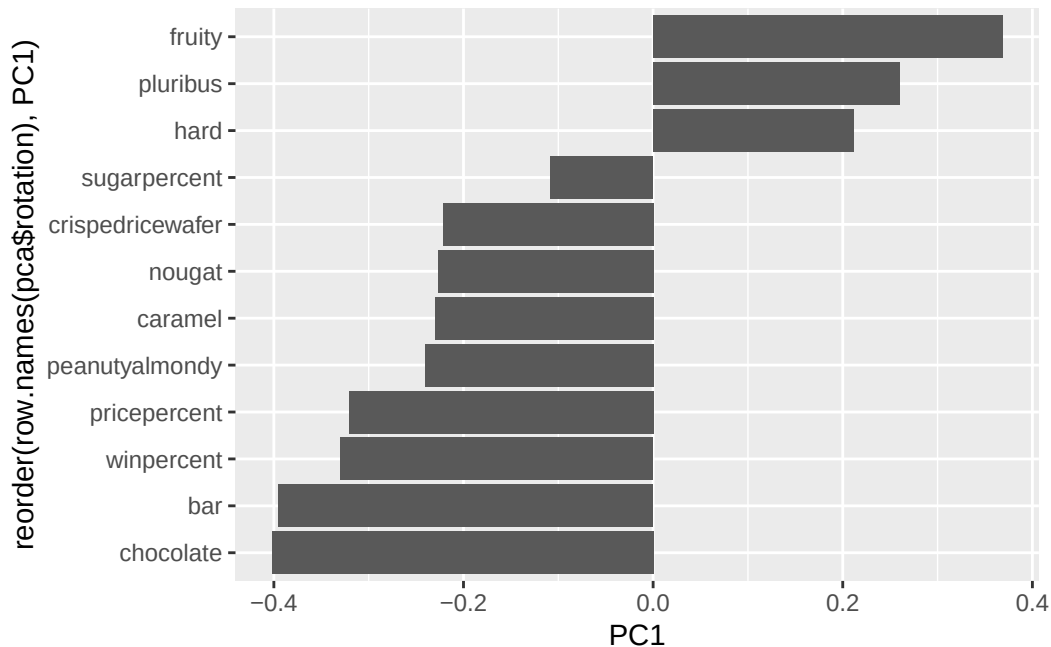
First major result figure is the “score plot” of PC1 vs PC2 - how different candy are related to each other on our PC axis

```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col=mycols) +
  geom_text_repel(size=3.3, col=mycols, max.overlaps=7)
```



The second major results figure from PCA is the so-called “loadings plot”

```
ggplot(pca$rotation)+
  aes(PC1,
    reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

In the positive direction fruity, pluribus, and hard are variables strongly associated in the positive direction. This makes sense as it seems like PC1 separated fruity candy to chocolate candy. This is also supported in the correlation plot as it shows how chocolate and fruity have a negative correlation in relation to each other.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A winning candy would be a chocolate based candy with features such as bar, peanuts, nougat, and caramel. As seen earlier chocolate candies have higher winpercent values as compared to fruit candies. The correlation analysis showed how chocolate is positively associated with winpercent. PCA showed that chocolate and bar candies are opposite from fruit candies and PC1 showing how chocolate candies is associated with a higher popularity.